

The defender of the multiverse documentation

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The Defender of the Multiverse

Introduction

"The Defender of the Multiverse" is a game that draws inspiration from the renowned bullet hell game "Vampire Survivors," with an added twist of tower defense mechanics. In this game, players are tasked with defending their crystal from swarms of enemies from various factions determined to destroy it.

Rules

In "The Defender of the Multiverse," the three enemy factions are modeled after those found in "StarCraft 2": Protoss, Zerg, and Terran. Each faction brings its unique set of four unit types, presenting different challenges and requiring specific strategies to defend against. As players rotate their perspective to switch between these factions, they must use their three types of spells effectively to protect their crystal from the diverse threats posed by the Protoss, Zerg, and Terran units. This dynamic adds depth and variety to the gameplay, echoing the rich lore and tactical gameplay of "StarCraft 2."

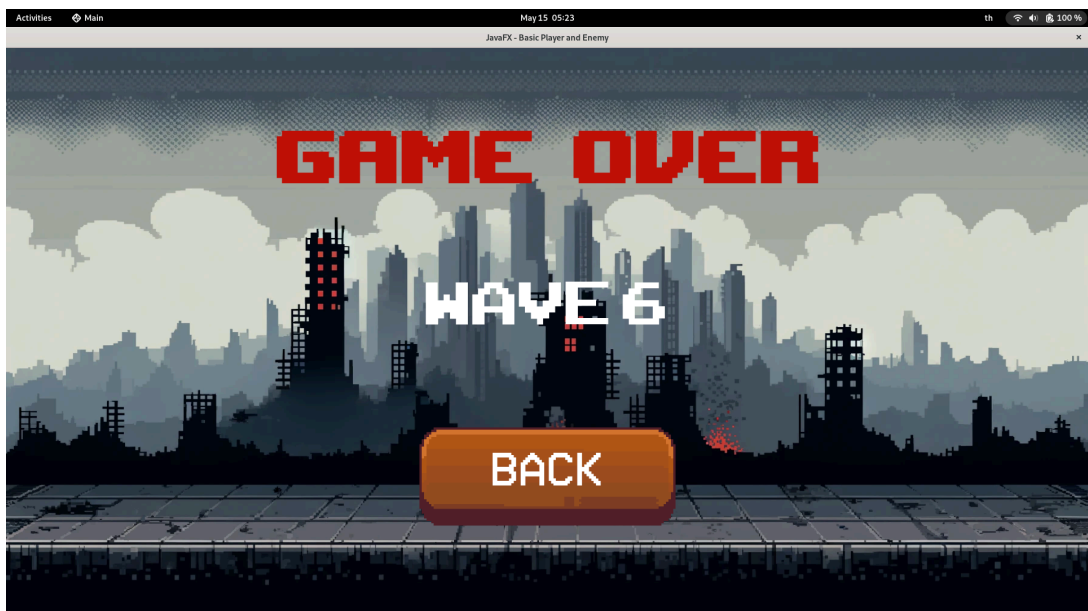
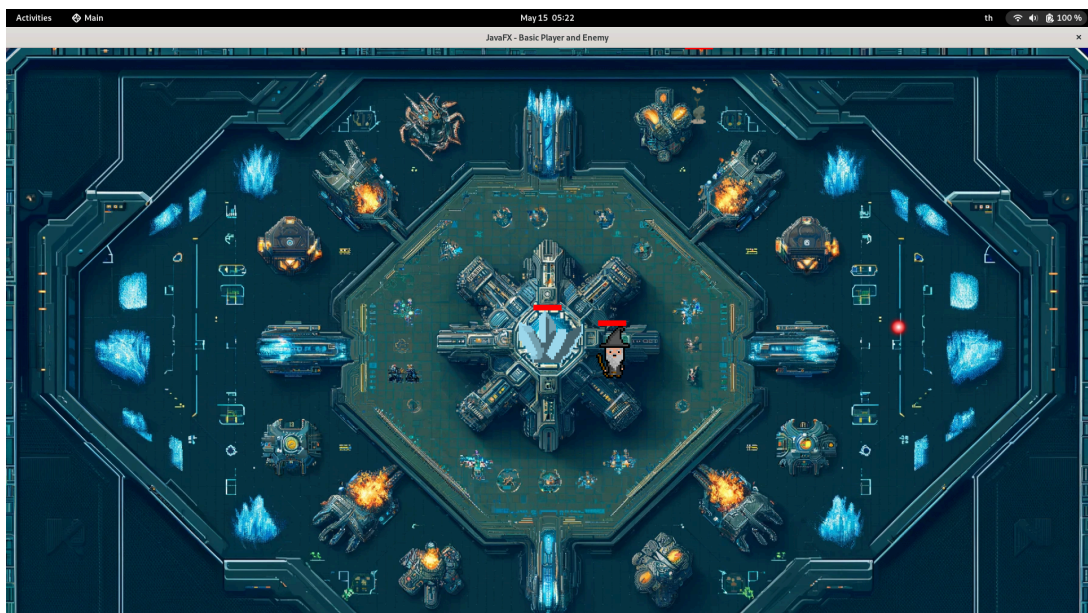
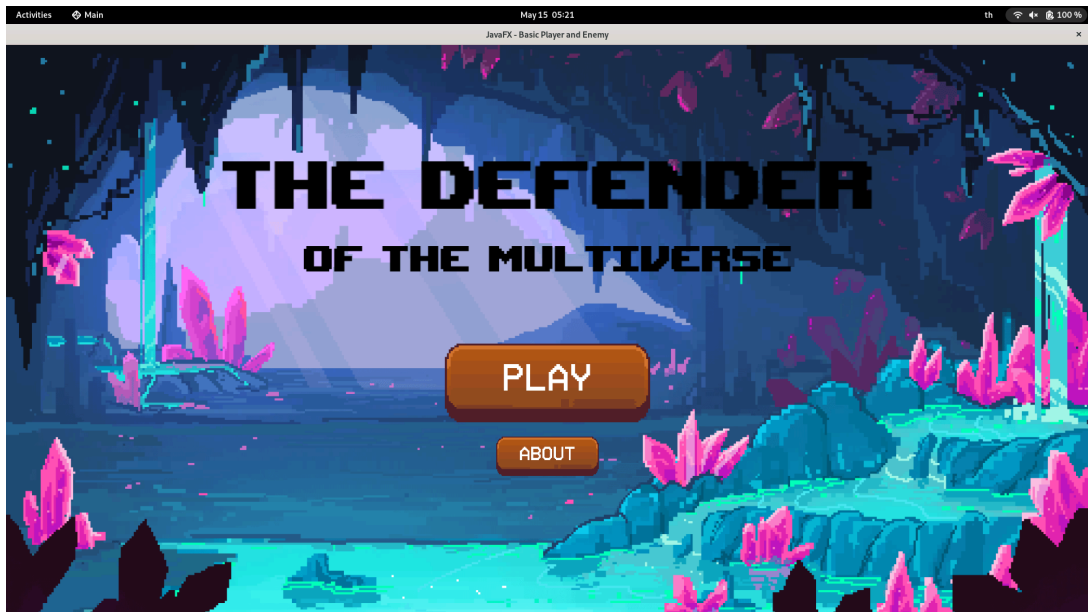
How to play

The goal of this game is to defend yourself and the crystal. The enemies will come in different races, skill, and stat. Surviving until the very end. Trying to get the highest score out of all.

The game lets you move freely around the crystal using WASD. You can press Q to switch to the previous race or E to switch to the next race. You can press the mouse to start shooting the basic attack. Or you can press 1,2,3,4 to toggle and click on the screen to cast a fireball, lighting orb, tornado, or gravitation field at the mouse position. You can further improve the spell by pressing shift or control when cast. Doing that will cause your spell to be enchant or make the spell casted in every dimension. All the spells (including the shift and control spell) have some cooldown.

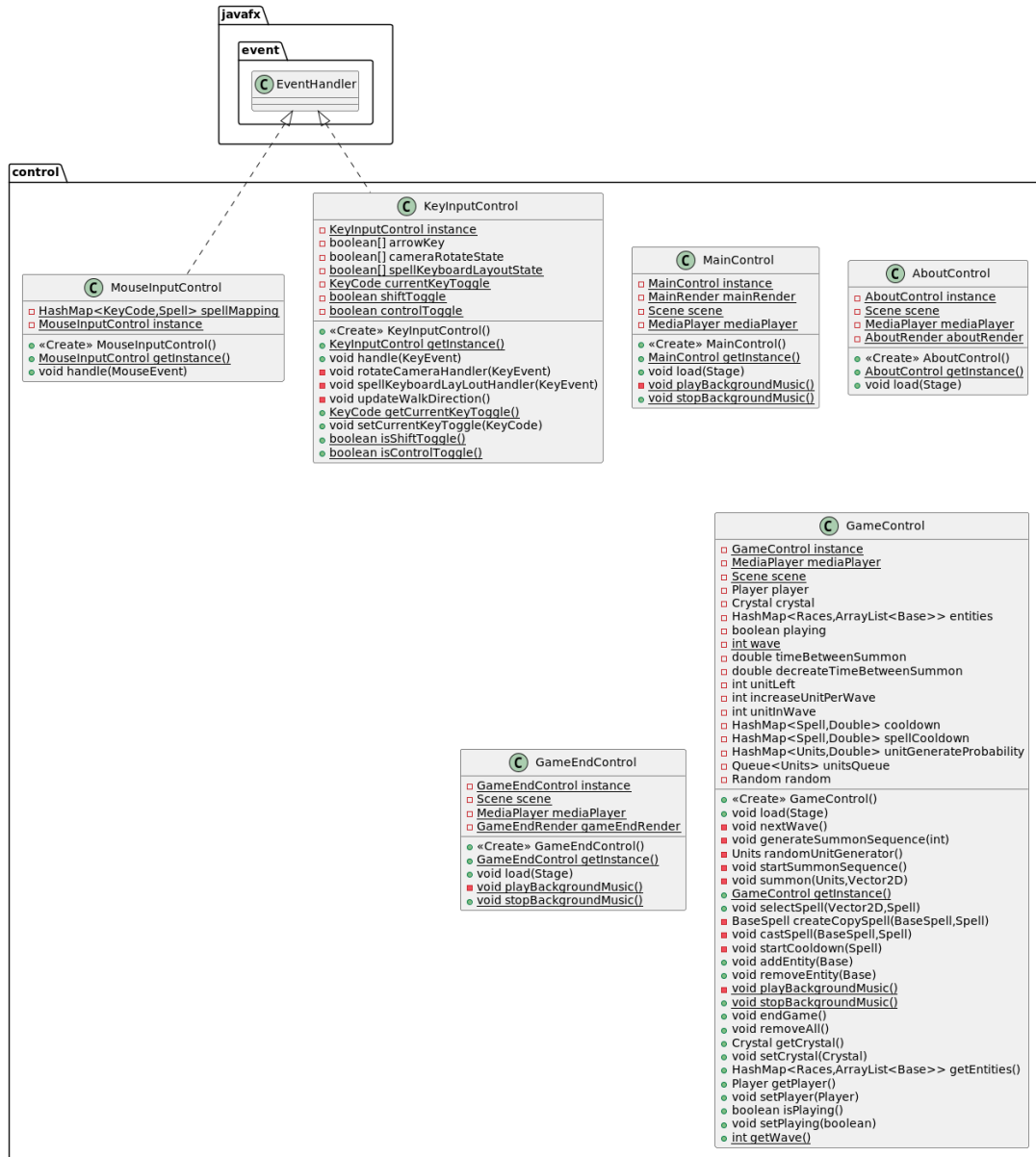
Current dimension that you are in will determine how you see another unit in another dimension. For example, if you are in the Terran dimension, you will see a red dot (Zerg unit) and blue dot (Protoss unit). You cannot see what it is, you can just see where it is on the map. The same case can be applied to the Zerg dimension and Protoss dimension, which will see terran as a green dot.

Example

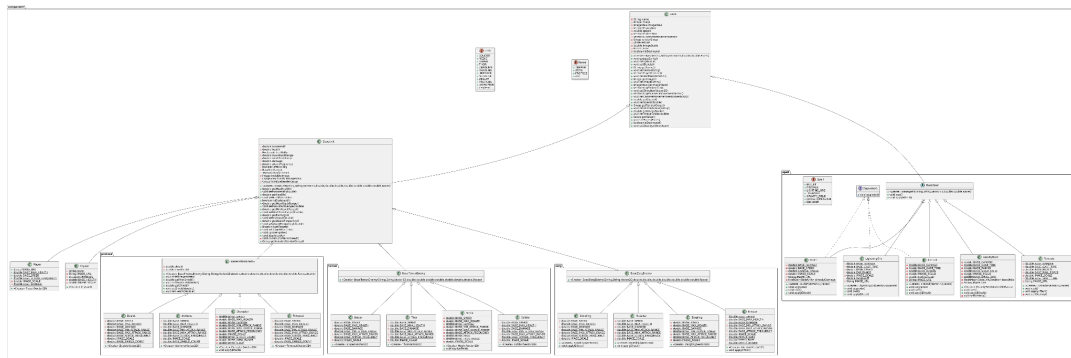


Class Diagram

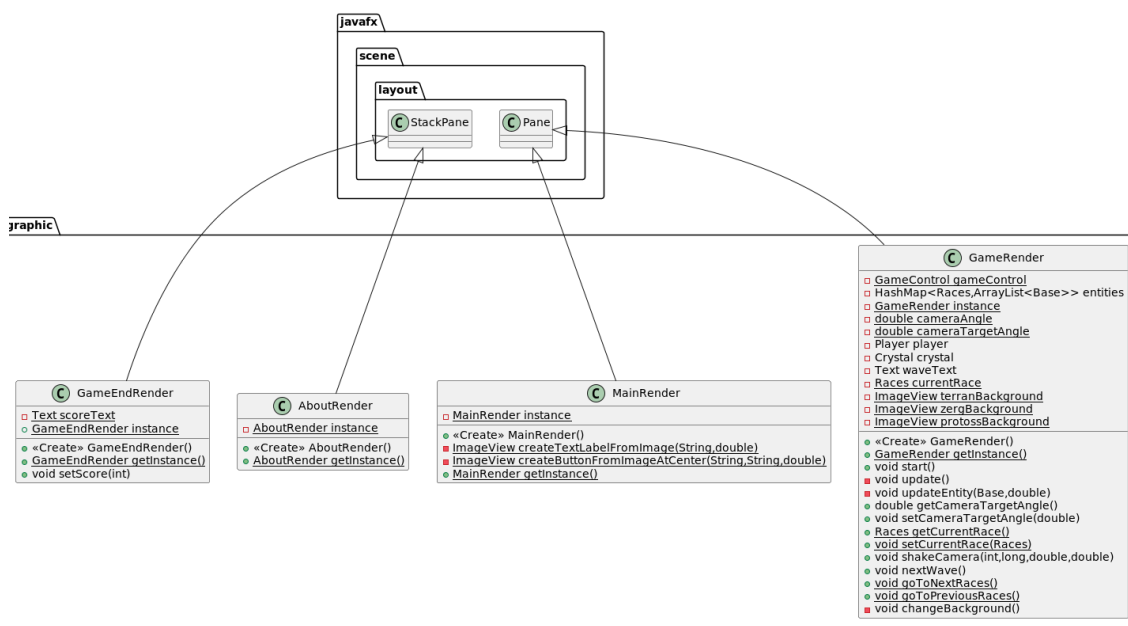
package control



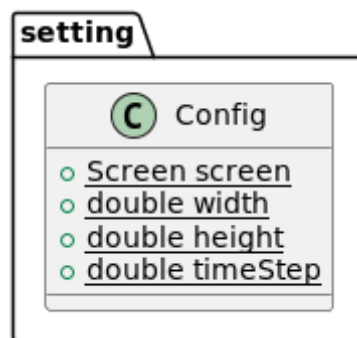
package component



package graphic



package setting



package util

util

Vector2D

- ❑ Random random
- Vector2D ZERO
- Vector2D UP
- Vector2D DOWN
- Vector2D RIGHT
- Vector2D LEFT
- Vector2D MID_SCREEN
- ❑ double x
- ❑ double y
- «Create» Vector2D()
- «Create» Vector2D(double,double)
- «Create» Vector2D(double,double,double)
- Vector2D add(Vector2D)
- Vector2D add(double,double)
- Vector2D subtract(Vector2D)
- Vector2D subtract(double,double)
- Vector2D multiply(double)
- Vector2D reverse()
- Vector2D rotateBy90(boolean)
- double getSize()
- Vector2D getNormalize()
- double getX()
- void setX(double)
- double getY()
- void setY(double)
- String toString()

Goto

- ❑ Stage stage
- ❑ Goto instance
- «Create» Goto()
- Goto getInstance()
- void gotoGame()
- void gotoMain()
- void gotoEndGame()
- void gotoAbout()
- void stopAllMusic()
- void setStage(Stage)

CreateButton

- ImageView createButtonFromImage(String,String,double)

1. package Component

1.1 class Base

- String name	Store unit's name
- Image image	Store unit's image
- ImageView imageView	Store unit's imageview
- Vector2D position	Store position of unit
- double speed	Store unit's speed
- Vector2D direction	Store unit's direction
- Vector2D cameraMovementVector	Store unit's movement vector that got generated by camera
- Group renderGroup	Store unit's render group (the group store imageView)
- Image invisibleImage	Store unit's image that got show when the main game isn't showing its race
- ImageView invisibleImageView	Store unit's ImageView for invisibleImage
- Group invisibleRenderGroup	Store unit's render group (the group store invisibleImageView)
- Circle redDot	For testing, not using in real game
- double imageScale	Store unit's image scale, for resizing image
- Races races	Store unit's race
- boolean isDestroyed	Store unit's destroyed state
- boolean renderAsMainRace	Store utility variable, for rendering
+ Base(String name, String imageUrl, Vector2D position, double speed, double imageScale, Races races)	Constructor for Base
+ void updateSprite(boolean renderAsMainRace)	Method for re rendering the image
+ void step(double time)	Method for calculate new position based on speed and direction
+ void selfDelete()	Method for delete itself
+ Getters/Setters for everything	

1.2 class BaseUnit extends Base

- double maxHealth	Store unit's max health
- double health	Store unit's current health
- Rectangle healthBar	Store unit's health bar
- double maxAttackRange	Store unit's maximum attack range
- double minAttackRange	Store unit's minimum attack range
- double damage	Store unit's attack damage
- double attackFrequency	Store unit's attack frequency (time/second)
- boolean attackFlag	Store unit's attack status (will change to true if unit's range is in between maximum attack range and minimum attack range, otherwise will be false)
- BaseUnit target	Store unit' target, the unit will try to move to the target until it is in attack range, and move back when they get too close. this field can be null.
- Thread attackThread	Store unit's attack thread.
+ BaseUnit(String name, String imageUrl, Vector2D position, double maxHealth, double speed, double imageScale, double minAttackRange, double maxAttackRange, double damage, double attackFrequency, Races races)	Constructor for BaseUnit
+ Getters/Setters for everything	

1.3 class Crystal extends BaseUnit

- String NAME	The name of crystal "Crystal"
- String IMAGE_URL	The image path of the crystal
- Vector2D POSITION	The initial position of the crystal, at the middle of the scene
- double BASE_HEALTH	The base health of the crystal
- double IMAGE_SCALE	The base image scale of the crystal
+ Crystal() {super(name, imageUrl, position,maxHealth,0,IMAGE_SCALE,	Constructor for Crystal

-1,-1, 0, 1,Races.ALL)	
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1.4 class Player extends BaseUnit

- <u>String IMAGE_URL</u>	The image url of the player
- <u>double BASE_MAX_HEALTH</u>	The base max health of the player
- <u>double BASE_SPEED</u>	The base movespeed of the player
- <u>double BASE_ATTACK_FREQUENCY</u>	The base attack frequency of the player
- <u>double IMAGE_SCALE</u>	The base image scale of the player
- <u>double BASE_DAMAGE</u>	The base damage of the player
+ Player(Vector2D position)	Constructor for Player

1.5 class Races

1.6 package Protoss

1.6.1 class BaseProtossEnemy extends BaseUnit

- double shield	Store unit's current shield, the shield act as a second health when shield is greater than 0
- double maxShield	Store unit's max shield
+ BaseProtossEnemy(String name, String imageUrl, Vector2D position, double maxHealth, double speed, double imageScale, double minAttackRange, double maxAttackRange, double damage, double attackChargeTime, Races races, double maxShield)	Constructor for BaseProtossEnemy
+ public void shieldRegenerate()	A function that create a thread for passive regeneration of the shield, the shield regenerate 1 unit every 1 second by default
+ Getters/Setters for everything	

1.6.2 class Archons extends BaseProtossEnemy

- <u>double BASE_SPEED</u>	The base speed of the archons
- <u>double BASE_MAX_HEALTH</u>	The base max health of the archons
- <u>double BASE_DAMAGE</u>	The base damage of the archons
- <u>double</u> <u>BASE_MIN_ATTACK_RANGE</u>	The base minimum attack range of the archons
- <u>double</u> <u>BASE_MAX_ATTACK_RANGE</u>	The base maximum attack range of the archons
- <u>double</u> <u>BASE_ATTACK_FREQUENCY</u>	The base attack frequency of the archons
- <u>double IMAGE_SCALE</u>	The base image scale of the archons
- <u>double BASE_SHIELD_COUNT</u>	The base shield count of the archons
+ Archons(Vector2D position)	Constructor for Archons

1.6.3 class Disruptor extends BaseProtossEnemy

- <u>double BASE_SPEED</u>	The base speed of the disruptor
- <u>double BASE_MAX_HEALTH</u>	The base max health of the disruptor
- <u>double BASE_DAMAGE</u>	The base damage of the disruptor
- <u>double</u> <u>BASE_MIN_ATTACK_RANGE</u>	The base minimum attack range of the disruptor
- <u>double</u> <u>BASE_MAX_ATTACK_RANGE</u>	The base maximum attack range of the disruptor
- <u>double</u> <u>BASE_ATTACK_FREQUENCY</u>	The base attack frequency of the disruptor
- <u>double IMAGE_SCALE</u>	The base image scale of the disruptor
- <u>double BASE_SHIELD_COUNT</u>	The base shield count of the disruptor
- <u>double BASE_AURA_RANGE</u>	The base aura range of the disruptor (this unit can slow player, if the player is in range)
- <u>double BASE_SLOW</u>	The base slow factor of the disruptor (this unit will slow player to certain ratio)
+ Disruptor(Vector2D position)	Constructor for Disruptor
+ void applyEffect()	Method for creating a thread that cast a spell that slow player

1.6.4 class Tempest extends BaseProtossEnemy

- <u>double BASE_SPEED</u>	The base speed of the tempest
- <u>double BASE_MAX_HEALTH</u>	The base max health of the tempest
- <u>double BASE_DAMAGE</u>	The base damage of the tempest
- <u>double</u> <u>BASE_MIN_ATTACK_RANGE</u>	The base minimum attack range of the tempest
- <u>double</u> <u>BASE_MAX_ATTACK_RANGE</u>	The base maximum attack range of the tempest
- <u>double</u> <u>BASE_ATTACK_FREQUENCY</u>	The base attack frequency of the tempest
- <u>double IMAGE_SCALE</u>	The base image scale of the tempest
- <u>double BASE_SHIELD_COUNT</u>	The base shield count of the tempest
+ Tempest(Vector2D position)	Constructor for Tempest
+ Getters/Setters for life	

1.6.5 class Zealot extends BaseProtossEnemy

- <u>double BASE_SPEED</u>	The base speed of the zealot
- <u>double BASE_MAX_HEALTH</u>	The base max health of the zealot
- <u>double BASE_DAMAGE</u>	The base damage of the zealot
- <u>double</u> <u>BASE_MIN_ATTACK_RANGE</u>	The base minimum attack range of the zealot
- <u>double</u> <u>BASE_MAX_ATTACK_RANGE</u>	The base maximum attack range of the zealot
- <u>double</u> <u>BASE_ATTACK_FREQUENCY</u>	The base attack frequency of the zealot
- <u>double IMAGE_SCALE</u>	The base image scale of the zealot
- <u>double BASE_SHIELD_COUNT</u>	The base shield count of the zealot
+ Zealot(Vector2D position)	Constructor for Zealot

1.7 package terran

1.7.1 class BaseTerranEnemy extends BaseUnit

+ BaseTerranEnemy(String name, String imageUrl, Vector2D position, double maxHealth, double speed, double imageScale, double minAttackRange, double maxAttackRange, double damage, double attackChargeTime, Races races)	Constructor for BaseTerranEnemy
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1.7.2 class Solider extends BaseTerranEnemy

- <u>double BASE_SPEED</u>	The base speed of the solider
- <u>double BASE_MAX_HEALTH</u>	The base max health of the solider
- <u>double BASE_DAMAGE</u>	The base damage of the solider
- <u>double BASE_MIN_ATTACK_RANGE</u>	The base minimum attack range of the solider
- <u>double BASE_MAX_ATTACK_RANGE</u>	The base maximum attack range of the solider
- <u>double BASE_ATTACK_FREQUENCY</u>	The base attack frequency of the solider
- <u>double IMAGE_SCALE</u>	The base image scale of the solider
+ Solider(Vector2D position)	Constructor for Solider

1.7.3 class Thor extends BaseTerranEnemy

- <u>double BASE_SPEED</u>	The base speed of the thor
- <u>double BASE_MAX_HEALTH</u>	The base max health of the thor
- <u>double BASE_DAMAGE</u>	The base damage of the thor
- <u>double BASE_MIN_ATTACK_RANGE</u>	The base minimum attack range of the thor
- <u>double BASE_MAX_ATTACK_RANGE</u>	The base maximum attack range of the thor
- <u>double BASE_ATTACK_FREQUENCY</u>	The base attack frequency of the thor

- <u>double IMAGE_SCALE</u>	The base image scale of the thor
+ Thor(Vector2D position)	Constructor for Thor

1.7.4 class Sniper extends BaseTerranEnemy

- <u>double BASE_SPEED</u>	The base speed of the sniper
- <u>double BASE_MAX_HEALTH</u>	The base max health of the sniper
- <u>double BASE_DAMAGE</u>	The base damage of the sniper
- <u>double</u> <u>BASE_MIN_ATTACK_RANGE</u>	The base minimum attack range of the sniper
- <u>double</u> <u>BASE_MAX_ATTACK_RANGE</u>	The base maximum attack range of the sniper
- <u>double</u> <u>BASE_ATTACK_FREQUENCY</u>	The base attack frequency of the sniper
- <u>double IMAGE_SCALE</u>	The base image scale of the sniper
+ Sniper(Vector2D position)	Constructor for sniper

1.7.5 class Medic extends BaseTerranEnemy

- <u>double BASE_SPEED</u>	The base speed of the sniper
- <u>double BASE_MAX_HEALTH</u>	The base max health of the sniper
- <u>double BASE_DAMAGE</u>	The base damage of the sniper
- <u>double</u> <u>BASE_MIN_ATTACK_RANGE</u>	The base minimum attack range of the sniper
- <u>double</u> <u>BASE_MAX_ATTACK_RANGE</u>	The base maximum attack range of the sniper
- <u>double</u> <u>BASE_ATTACK_FREQUENCY</u>	The base attack frequency of the sniper
- <u>double IMAGE_SCALE</u>	The base image scale of the sniper
- <u>double BASE_HEAL_RANGE</u>	The base heal range for medic (this unit will heal all Terran unit within range)
- <u>double BASE_HEAL</u>	The base heal for medic (this unit will heal all Terran unit with this amount of heal)
+ Medic(Vector2D position)	Constructor for Medic

+ void applyEffect()	Method for starting the thread that heal every Terran unit within range, all unit will got +100% heal from BASE_HEAL except for thor (+150%)
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1.8 package Zerg

1.8.1 class BaseZergEnemy extends BaseUnit

+ BaseZergEnemy(String name, String imageUrl, Vector2D position, double maxHealth, double speed, double imageScale, double minAttackRange, double maxAttackRange, double damage, double attackFrequency, Races races)	Constructor for BaseZergEnemy
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1.8.2 class Zergling extends BaseZergEnemy

- <u>double BASE_SPEED</u>	The base speed of the zergling
- <u>double BASE_MAX_HEALTH</u>	The base max health of the zergling
- <u>double BASE_DAMAGE</u>	The base damage of the zergling
- <u>double BASE_MIN_ATTACK_RANGE</u>	The base minimum attack range of the zergling
- <u>double BASE_MAX_ATTACK_RANGE</u>	The base maximum attack range of the zergling
- <u>double BASE_ATTACK_FREQUENCY</u>	The base attack frequency of the zergling
- <u>double IMAGE_SCALE</u>	The base image scale of the zergling
+ Zergling(Vector2D position)	Constructor for Zergling

1.8.3 class Baneling extends BaseZergEnemy

- <u>double BASE_SPEED</u>	The base speed of the baneling
- <u>double BASE_MAX_HEALTH</u>	The base max health of the baneling
- <u>double BASE_DAMAGE</u>	The base damage of the baneling
- <u>double BASE_MIN_ATTACK_RANGE</u>	The base minimum attack range of the baneling

- <u>double</u> <u>BASE_MAX_ATTACK_RANGE</u>	The base maximum attack range of the baneling
- <u>double</u> <u>BASE_ATTACK_FREQUENCY</u>	The base attack frequency of the baneling
- <u>double</u> <u>IMAGE_SCALE</u>	The base image scale of the baneling
+ Baneling(Vector2D position)	Constructor for Zergling
+ applyEffect()	Method for starting the thread detects if the unit is within range of attack. If it in, it will explode. Deal some damage and kill itself.

1.8.4 class Infestor extends BaseZergEnemy

- <u>double</u> <u>BASE_SPEED</u>	The base speed of the infestor
- <u>double</u> <u>BASE_MAX_HEALTH</u>	The base max health of the infestor
- <u>double</u> <u>BASE_DAMAGE</u>	The base damage of the infestor
- <u>double</u> <u>BASE_MIN_ATTACK_RANGE</u>	The base minimum attack range of the infestor
- <u>double</u> <u>BASE_MAX_ATTACK_RANGE</u>	The base maximum attack range of the infestor
- <u>double</u> <u>BASE_ATTACK_FREQUENCY</u>	The base attack frequency of the infestor
- <u>double</u> <u>IMAGE_SCALE</u>	The base image scale of the infestor
- <u>double</u> <u>ATTACK_BUFF</u>	The base attack buff ratio (this unit can buff other Zerg unit)
- <u>double</u> <u>SPEED_BUFF</u>	The base speed buff ratio (this unit can buff other Zerg unit)
- <u>double</u> <u>SPEED_DEBUFF</u>	The base speed debuff ratio (this unit can slow player by a certain amount of ratio)
+ Infestor(Vector2D position)	Constructor for Infestor
+ applyEffect()	Method for starting the thread that apply all 3 effect to random zerg unit and player

1.8.5 class Mutalisk extends BaseZergEnemy

- <u>double</u> <u>BASE_SPEED</u>	The base speed of the mutalisk
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- <u>double BASE_MAX_HEALTH</u>	The base max health of the mutalisk
- <u>double BASE_DAMAGE</u>	The base damage of the mutalisk
- <u>double</u> <u>BASE_MIN_ATTACK_RANGE</u>	The base minimum attack range of the mutalisk
- <u>double</u> <u>BASE_MAX_ATTACK_RANGE</u>	The base maximum attack range of the mutalisk
- <u>double</u> <u>BASE_ATTACK_FREQUENCY</u>	The base attack frequency of the mutalisk
- <u>double IMAGE_SCALE</u>	The base image scale of the mutalisk
+ Mutalisk(Vector2D position)	Constructor for Mutalisk
+ applyEffect()	Method for starting the thread that constantly spawn new zergling every certain amount of time

1.9 package Spell

1.9.1 abstract class BaseSpell extends Base

+ BaseSpell(String name, String imageUrl, Vector2D position, double speed, double imageScale, Races races)	Constructor for BaseSpell
+ void cast() {}	Abstract method for cast the spell
+ void applyEffect() {}	Abstract method for applying the effect to unit

1.9.2 enum Spell

BULLET,
 FIREBALL,
 LIGHTING_ORB,
 TORNADO,
 GRAVITY_FIELD,
 CROSS_DIMENSION,
 ENCHANT

1.9.3 interface Upgradable

+ void upgrade()	Method for upgrading the spell
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1.9.4 class Bullet extends BaseSpell implements Upgradable

- <u>double BASE_DAMAGE</u>	The base damage of bullet
- <u>double BASE_CAST_TIME</u>	The base cast time of the bullet
- <u>double BASE_RADIUS</u>	The base radius of the bullet
- <u>double IMAGE_SCALE</u>	The base image scale of the bullet
- <u>String IMAGE_URL</u>	The base image url of the bullet
- HashSet<BaseUnit> alreadyDamage	Store all the units that have been damaged by this bullet so that it will not be damaged multiple times.
+ Bullet(Vector2D position, Races races)	Constructor for Bullet
+ void upgrade()	Method that change speed to 1000
+ void cast()	Method for generating an animation and play.
+ void applyEffect()	Method for applying the damage when the animation finished

1.9.5 class Fireball extends BaseSpell implements Upgradable

- <u>double BASE_DAMAGE</u>	The base damage of the fireball
- <u>double BASE_CAST_TIME</u>	The base cast time of the fireball
- <u>double BASE_RADIUS</u>	The base radius of the fireball
- <u>double IMAGE_SCALE</u>	The base image scale of the fireball
- <u>String IMAGE_URL</u>	The base image url of the fireball
- <u>double buffDamage</u>	Store spell's damage multiplier (set to 1 by default, will be 5 if upgraded)
+ Fireball(Vector2D position, Races races)	Constructor for Fireball
+ void upgrade()	Method that change buffDamage to 5
+ void cast()	Method for generating an animation and play.
+ void applyEffect()	Method for applying the damage when the animation finished

1.9.6 class GravityField extends BaseSpell

- <u>double BASE_DAMAGE</u>	The base damage of gravity field
- <u>double BASE_CAST_TIME</u>	The base cast time of gravity field
- <u>double BASE_RADIUS</u>	The base radius of gravity field
- <u>double BASE_SLOW_RATIO</u>	The base slow ratio for gravity field
- <u>double IMAGE_SCALE</u>	The base image scale of gravity field
- <u>double BASE_SPELL_TIME</u>	The base spell active time of gravity field
- <u>String IMAGE_URL</u>	The base image url of gravity field
+ GravityField(Vector2D position, Races races)	Constructor for Fireball
+ void cast()	Method for generating an animation and play.
+ void applyEffect()	Method for applying the damage when the animation finished
- selfDestroy()	Method that create thread for self destroy in BASE_SPELL_TIME

1.9.7 class LightningOrb extends BaseSpell implements Upgradable

- <u>double BASE_DAMAGE</u>	The base damage of lighting orb
- <u>double BASE_CAST_TIME</u>	The base cast time of the lighting orb
- <u>double ZAP_RANGE</u>	The base zap range of the lighting orb (this spell will damage every enemy nearby every certain amount of time)
- <u>double IMAGE_SCALE</u>	The base image scale of the lighting orb
- <u>String IMAGE_URL</u>	The base image url of the lighting orb
- <u>double rangeBuff</u>	Store spell's range multiplier (set to 1 by default, will be 2.5 if upgraded)
+ LightningOrb(Vector2D position, Races races)	Constructor for LightningOrb
+ void upgrade(Empower empower)	Method that change rangeBuff to 5
+ void cast()	Method for generating an animation and play
+ void applyEffect()	Method for applying the damage when the animation finished

1.9.8 class Tornado extends BaseSpell implements Upgradable

- <u>double BASE_DAMAGE</u>	The base damage of tornado
- <u>double BASE_CAST_TIME</u>	The base cast time of tornado
- <u>double BASE_RADIUS</u>	The base effect radius of tornado
- <u>double IMAGE_SCALE</u>	The base image scale of tornado
- <u>double BASE_PULL_FORCE</u>	The base pull force of tornado (this spell will try to pull unit into center, this parameter adjust the power of the pull)
- <u>String IMAGE_URL</u>	The base image url of tornado
+ Tornado(Vector2D position, Races races)	Constructor for LightingOrb
+ void cast()	Method for generating an animation and play
+ void applyEffect()	Method for applying the pull when the animation finished
- selfDestroy()	Method that create thread for self destroy in certain amount of time

2. package control

2.1 class AboutControl

- <u>AboutControl instance</u>	instance of AboutControl
- <u>Scene scene</u>	Store scene created by aboutRender
- <u>MediaPlayer mediaPlayer</u>	Store background music for about page
- <u>AboutRender aboutRender</u>	Store AboutRender's instance
+ AboutControl()	Constructor for AboutControl
+ <u>AboutControl getInstance()</u>	Method for getting an instance
+ load(Stage stage)	Method for setting the stage to this scene

2.2 class GameControl

- <u>GameControl instance</u>	instance of GameControl
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- <u>MediaPlayer mediaPlayer</u>	Store background music for game page
- <u>Scene scene</u>	Store scene created by GameRender
- Player player	Store player instance
- Crystal crystal	Store crystal instance
- HashMap<Races, ArrayList<Base>> entities	Store all units that is currently in game
- boolean playing	Store game state
- <u>int wave</u>	Store the current wave number
- double timeBetweenSummon	Store the time between each unit spawn
- double decreaseTimeBetweenSummon	Store the decrease time in summoning time
- int unitLeft	Store the current unit left in game
- int increaseUnitPerWave	Store the increasing number of unit per wave
- int unitInWave	Store the number of unit in this wave that will be spawn
- HashMap<Spell, Double> cooldown	Store the real time cooldown for spell
- HashMap<Spell, Double> spellCooldown	Store the base cooldown time for each spell
- HashMap<Spell, Double> unitGenerateProbability	Store the probability of generating certain unit
- Queue<Units> unitsQueue	Store the spawning queue
- Random random	Store the Random instance
+ GameControl()	Constructor for GameControl
+ void load()	Load the game scene to the stage
+ void nextWave()	Go to the next wave, tell the GameRender class to go to the nextwave too. Increase the number of unit in wave, and call generateSummonSequence
- void generateSummonSequence(int unitNum)	Create a summoning queue for unit. The total number of units will be equal to unitNum.
- Units randomUnitGenerator()	Create random unit base on probability
- void startSummonSequence()	Start summoning each unit, one by one. Each one will be separate by a certain amount of time
- void summon(Units unit, Vector2D	Summon the unit at position

position)	
+ <u>GameControl getInstance()</u>	get GameControl instance
+ void selectSpell(Vector2D mousePosition, Spell spell)	selected the spell to be cast. If not on cooldown, the process will be pass to castSpell
- BaseSpell createCopySpell(BaseSpell spell, Spell spellType)	Create an exact copy of certain spell
- void castSpell(BaseSpell spell, Spell spellType)	Casting the spell then call startCooldown for that spell
- void startCooldown(Spell spellType)	Create a thread that account for cooldown for that spell
+ void addEntity(Base entity)	Add entity to game, call GameRender for adding the renderGroup
+ void removeEntity(Base entity)	Remove entity from game, call GameRender for removing the renderGroup
- void playBackgroundMusic()	Play the background music
+ void stopBackgroundMusic()	Stop the background music
+ void endGame()	Set isPlaying to false, then remove all unit and end game
+ void removeAll	Remove everything from the game
+ Getters/Setters for everything	

2.3 class GameEndControl

- <u>GameEndControl instance</u>	instance of GameEndControl
- <u>Scene scene</u>	Store scene of GameEndRender
- <u>MediaPlayer mediaPlayer</u>	Store game end background music
- <u>GameEndRender gameEndRender</u>	Store GameEndRender's instance
+ GameEndControl()	Constructor for GameEndControl
+ <u>GameEndControl getInstance()</u>	get GameEndControl's instance
+ void load()	Set the stage to end game page

2.4 class KeyInputControl implements EventHandler<KeyEvent>

- <u>KeyInputControl instance</u>	KeyInputControl's instance
- boolean[] arrowKey	Store WASD key state (for walking)
- boolean[] cameraRotateState	Store QE key state (for rotating the camera)
- <u>boolean[] spellKeyboardLayoutState</u>	Store 1 2 3 4 key state (for casting a spell)
+ KeyInputControl()	Constructor for KeyInputControl
+ <u>KeyInputControl getInstance()</u>	Get KeyInputControl's instance
+ void handle(KeyEvent event)	Method that handle all key event, the rerouting the event to certain function
+ void rotateCameraHandler(KeyEvent event)	Method that order the rotation of the camera and prevent player from constantly rotate the camera
- spellKeyboardLeyLoutHandler(KeyEvent event)	Method that casting the spell based on the key and keyevent
- void updateWalkDirection()	Method for updating the walk direction based on WASD

2.5 class MainControl

- <u>MainControl instance</u>	instance of MainControl
- <u>MainRender mainRender</u>	Store MainRender's instance
- <u>Scene scene</u>	Store main page scene
- <u>MediaPlayer mediaPlayer</u>	Store main page background music
+ MainControl()	Constructor for MainControl
+ <u>MainControl getInstance()</u>	Get MainControl's instance
+ void load()	Set the stage to this scene
- <u>void playBackgroundMusic()</u>	Play the background music
+ <u>video stopBackgroundMusic()</u>	Stop the background music

2.6 class MouseInputControl implements EventHandler<MouseEvent>

- <u>MouseInputControl instance</u>	instance of MouseInputControl
- <u>HashMap<KeyCode, Spell> spellMapping</u>	Store which key map to what spell
+ MouseInputControl()	Constructor for MouseInputControl
+ <u>MouseInputControl getInstance()</u>	Get MouseInputControl' instance
+ void handle(MouseEvent event)	Event handler for mouse event, use for trigger spell casting

3. package Graphic

3.1 class AboutRender extends StackPane

- <u>AboutRender instance</u>	instance of AboutRender
+ <u>AboutRender()</u>	Constructor for AboutRender, create every component in this stackpane
- <u>AboutRender getInstance()</u>	Get AboutRender's instance

3.1 class GameEndRender extends StackPane

- <u>GameEndRender instance</u>	instance of AboutRender
- <u>Text scoreText</u>	Text for displaying score
+ GameEndRender()	Constructor for AboutRender, create every component in this stackpane
- <u>GameEndRender getInstance()</u>	Get AboutRender's instance
- void setScore	Set the score for text

3.1 class GameRender extends Pane

- <u>GameControl gameControl</u>	instance of GameControl
- <u>HashMap<Races, ArrayList<Base>></u>	Store all entities in game

<u>entities</u>	
- <u>GameRender instance</u>	Store GameRender's instance
- <u>double cameraAngle</u>	Store current camera angle
- <u>double cameraTargetAngle</u>	Store the target camera angle
- <u>Player player</u>	Store player's instance
- <u>Crystal crystal</u>	Store crystal's instance
- <u>Text waveText</u>	Store the wave text showed at the end of the wave.
- <u>Races currentRace</u>	Store current display race
- <u>ImageView terranBackground</u>	Store Terran dimension background
- <u>ImageView zergBackground</u>	Store Zerg dimension background
- <u>ImageView protossBackground</u>	Store Protoss dimension background
+ GameRender()	Constructor for GameRender. Initialize some component in Pane
+ <u>GameRender getInstance()</u>	Get GameRender's instance
+ void start()	Initialize some component in Pane and create thread for re rendering the scene
+ void update()	Method that make all component re rendering
+ void updateEntity(Base entity, double rotateAngle)	Method for updating each entity. This got called by update function
+ void shakeCamera(int numShakes, long duration, double shakeStrengthIn, double decayShakeStrength)	Method for simulating a camera shaking effect
+ void nextWave()	Method that got called when the wave is over. Show next wave text.
+ <u>void goToNextRaces()</u>	Go to the next race
+ <u>void goToPreviousRaces()</u>	Go to the previous race
+ <u>void changeBackground()</u>	Method that change background according to current race
+ Getters/Setters for everything	

3.2 class MainRender extends Pane

- <u>MainRender instance</u>	instance of MainRender
+ <u>MainRender()</u>	Constructor for MainRender, create every component in this Pane
- <u>ImageView</u> <u>createTextLabelFromImage(String imagePath, double sizeCompareToScreenWidth)</u>	Method for creating a text from image
- <u>ImageView</u> <u>createButtonFromImageAtCenter(String imagePath1, String imagePath2, double sizeCompareToScreenWidth)</u>	Method for creating a button from image
+ <u>MainRender getInstance()</u>	Get MainRender's instance

4. package setting

4.1 class Config

+ <u>Screen screen</u>	instance of screen, the primary one.
+ <u>double width</u>	screen's width
+ <u>double height</u>	screen's height
+ <u>double timeStep</u>	the time between re rendering (in second)

5. package util

5.1 class CreateButton

+ <u>ImageView</u> <u>createButtonFromImage(String imagePath1, String imagePath2, double sizeCompareToScreenWidth)</u>	Create a button from an image, as the name suggest.
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5.2 class Goto

- <u>Stage stage</u>	Store an application main stage
- <u>Goto instance</u>	instance of Goto
+ <u>gotoGame()</u>	Go to game page, call load() in GameControl

+ gotoMain()	Go to main page, call load() in MainControl
+ gotoEndGame()	Go to end game page, call load() in GameEndControl
+ gotoAbout()	Go to about page, call load() in AboutControl
- stopAllMusic	Stop all music

5.3 class Vector2D

- <u>Random random</u>	Random's instance
- <u>Vector2D ZERO</u>	A (0,0) vector
+ <u>Vector2D UP</u>	A (0,-1) vector
+ <u>Vector2D DOWN</u>	A (0,1) vector
+ <u>Vector2D RIGHT</u>	A (1,0) vector
+ <u>Vector2D LEFT</u>	A (-1,0) vector
- <u>Vector2D MID_SCREEN</u>	A vector at the middle of the screen
- double x	x coordination of vector
- double y	y coordination of vector
+ Vector2D()	Create a random vector located inside a screen width and height
+ Vector2D(double x, double y)	Create a vector (x,y)
+ Vector2D(double x, double y, double size)	Create a vector (x * size,y * size)
+ Vector2D add(Vector2D in)	Method for creating a new vector that is already added by in vector
+ Vector2D add(double x, double y)	Method for creating a new vector (this.x + x, this.y + y)
+ Vector2D subtract(Vector2D in)	Method for creating a new vector that is already subtracted by in vector
+ Vector2D subtract(double x, double y)	Method for creating a new vector (this.x - x, this.y - y)
+ Vector2D multiply(double mul)	Method for creating a new vector that got multiply by mul
+ Vector2D rotateBy90(boolean clockwise)	Rotated a vector by 90 or -90

+ double getSize()	Get the vector size
+ Vector2D getNormalize()	Return a normalize version of this vector
+ String toString()	Return a string representation of this vector
+ Getters/Setters for everything	